

Daily Research Digest — 2026-04-13

Topics: galaxy clusters, galaxies, gravitational lensing, dark matter.

Synthesis

Executive overview

Recent preprints in the arXiv repository have introduced a variety of new insights into galaxy clusters, galaxies, gravitational lensing, and dark matter. Notably, there is significant progress in understanding ultra-faint satellite galaxies around the Milky Way, as well as advancements in modeling star formation rates and gas dynamics within various types of galaxies. Additionally, methodological improvements for analyzing Lyman-alpha forest data have been proposed to enhance our ability to constrain cosmological models.

Galaxy clusters

There are no preprints directly focused on galaxy clusters in this list. This indicates a potential gap in recent submissions regarding the dynamics and evolution of these large-scale structures.

Galaxies

Recent work has delved into understanding the missing ultra-faint satellite galaxies around the Milky Way, suggesting that there are fewer satellites at very low luminosities than expected due to tides and dark matter stripping. Another study explores extreme emission-line galaxies (EELGs) in the local universe, which serve as analogs for reionization-era galaxies, highlighting the importance of high-redshift star formation studies. Furthermore, research on the molecular Kennicutt-Schmidt relation reveals that higher-J CO transitions indicate more linear correlations between gas and star formation rates, suggesting denser gas is directly linked to ongoing star formation. Lastly, [CII] emissions from high-redshift quiescent galaxies provide insights into molecular gas fractions and dust heating mechanisms beyond standard stellar radiation.

Gravitational lensing

There are no preprints focusing specifically on gravitational lensing in this list, indicating a potential area for future research or fewer recent developments compared to other areas.

Dark matter

One paper explores the effective field theory (EFT) applied to the Lyman-alpha forest flux power spectrum, proposing methods to reduce computational costs and enhance model compression through Fisher matrix formalism. This work aims to better understand small-scale structure formation and its implications for dark matter models.

Cross-cutting themes

Several papers highlight methodological advancements in analyzing galaxy data, such as using machine learning techniques (k-means) for classifying extreme emission-line galaxies and employing Fisher matrices to improve model compression for cosmological probes. Additionally, there is a consistent focus on connecting observational data with theoretical predictions through sophisticated modeling of star formation rates, gas dynamics, and the influence of dark matter.

Papers to read first

- ****The missing ultra-faint satellites of the Milky Way****: This paper provides important insights into the population of faint satellite galaxies around the Milky Way, crucial for understanding galaxy evolution.
- ****The k-MENDEL sample of local analogs to reionization galaxies. Spectral identification of EELGs and properties of green peas in DESI****: This study offers a robust classification method for extreme emission-line galaxies, critical for probing early star formation processes.
- ****Constraining the Molecular Kennicutt-Schmidt Relation with Multi-Transition CO Observations of Nearby Galaxies****: The analysis provides detailed information on how different transitions of CO relate to star formation rates in nearby galaxies, shedding light on gas dynamics and stellar evolution.

Open questions and follow-ups

Despite advancements, there remain several open areas. For example, the exact nature and number of ultra-faint dwarf satellites is still unclear and requires more observational data. Additionally, understanding the precise mechanisms driving dust heating in high-redshift quiescent galaxies remains a key challenge. Further work on methodological improvements for analyzing large datasets, such as those from Lyman-alpha forests, will be essential to constrain cosmological parameters accurately.

Paper Summaries and Figures

1. The missing ultra-faint satellites of the Milky Way

Focus: galaxies, dark matter

Authors: Isabel M. E. Santos-Santos, Carlos S. Frenk, Julio F. Navarro

Published: 2026-04-10T17:57:28+00:00

PDF: <https://arxiv.org/pdf/2604.09539v1>

Summary: We combine the highest resolution N-body simulation of a $\sim 10^{12} M_{\odot}$ CDM halo (Aquarius-A) with the {\sc GALFORM} galaxy formation semianalytic model to study the full satellite population expected in a MW-like galaxy. The model assumes that galaxies only form in subhalos whose peak circular velocity exceeds the H-cooling threshold, all of which are well resolved in the simulation. The number of luminous subhalos ever accreted into the main halo is thus well defined, and implies that the total number of MW satellites, down to arbitrarily low luminosity, should not exceed a few hundred. The model tracks satellites even after their halos cease to be resolved ("orphan" galaxies), and includes a novel treatment of dark matter and stellar tidal stripping which takes into account that all CDM subhalos survive until the present because of their cuspy inner density profiles. After accounting for tides, our results match well the massive end of the observed MW satellite mass function and predict that a large number of ultra-faint dwarfs are missing from the current MW satellite census. The missing UFDs are predicted to avoid the innermost regions of the host, and to have properties that overlap with those of the many ultra-faint compact MW satellites (UFCs) discovered recently, with properties intermediate between globular clusters and dwarf galaxies. Our results suggest that many UFC systems are dark matter-dominated dwarfs with velocity dispersions between $1 - 3 \text{ km/s}$, which have survived disruption because they reside in the dense cusp of CDM subhalos. UFCs should have mean densities of order $10^{10} - 10^{11} M_{\odot}/\text{kpc}^3$, higher than those of more extended ultra-faint systems. If confirmed, our results would provide support for the cuspy nature of CDM dark matter halos and for the hydrogen-cooling threshold for galaxy formation.

2. The k-MENDEL sample of local analogs to reionization galaxies. Spectral identification of EELGs and properties of green peas in DESI

Focus: galaxies

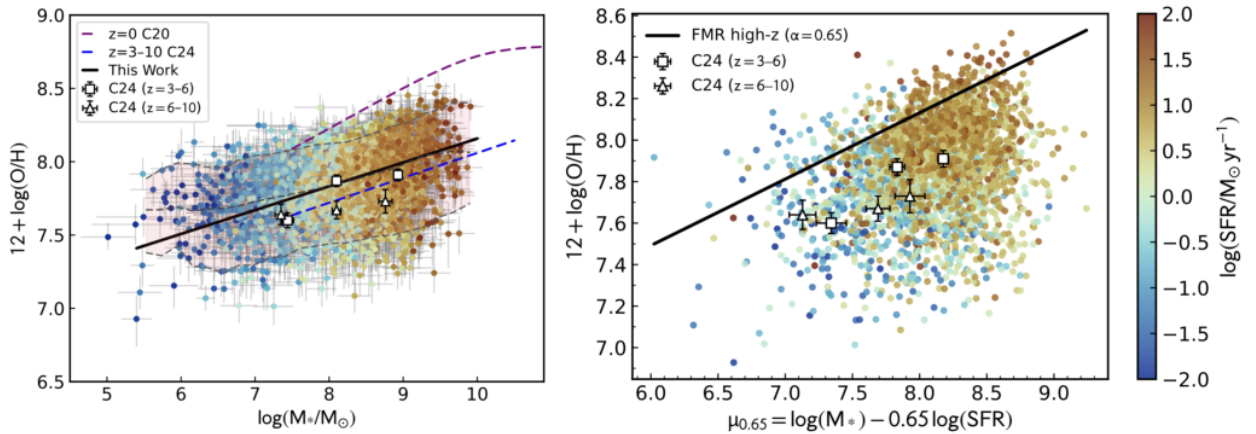
Authors: L. Bonatto, R. Amorín, A. Giménez-Alcázar, J. A. Fernández-Ontiveros, A. Hernán-Caballero, S. Suárez, J. M. Vílchez, E. Pérez-Montero, M. Llerena, J. Sánchez Almeida

Published: 2026-04-10T17:37:50+00:00

PDF: <https://arxiv.org/pdf/2604.09516v1>

Summary: Low-mass galaxies with intense starbursts exhibit spectra dominated by extreme nebular emission and faint stellar continua. These extreme emission-line galaxies (EELGs) are key laboratories to study star formation, feedback, and ionizing photon escape in low-metallicity environments. We exploit the DESI survey to assemble the k-Means of Extreme Nebulae from DESI outLiers (k-MENDEL), a statistically robust sample of $\sim 16,000$ EELGs at $0.01 < z < 0.96$ selected via automatic k-means classification. Using SED fitting and Te-based metallicities, we characterize EELGs including "blueberry" and "green pea" galaxies, spanning stellar masses of 10^6 - 10^{10} Msun and SFRs of 0.1-100 Msun/yr. k-MENDEL extends previous SDSS samples toward higher redshifts and lower metallicities ($12+\log(\text{O}/\text{H}) \sim 7.0$ -8.5). EELGs lie systematically above the star-forming main sequence, with sSFRs up to $\sim 100 \text{ Gyr}^{-1}$. They follow a shallower mass-metallicity relation offset by 0.3-0.5 dex from local relations, closely resembling young galaxies observed with JWST at $z > 3$ -10. The large intrinsic metallicity scatter, even after projecting along the fundamental metallicity relation, indicates strong departures from simple "bathtub" models, suggesting massive inflows of metal-poor gas followed by strong feedback. While $\sim 6\%$ of the sample shows AGN-like signatures, the most extreme star-forming systems reach high ionization ($\text{O32} \sim 5$ -60) comparable to confirmed Lyman-continuum emitters. Our results support the interpretation of EELGs as short-lived, non-equilibrium phases in the evolution of low-mass galaxies and highlight their importance as nearby analogs of galaxies likely driving cosmic reionization (Abridged).

Selected figures



3. Analytic compression of the effective field theory of the Lyman-alpha forest

Focus: dark matter

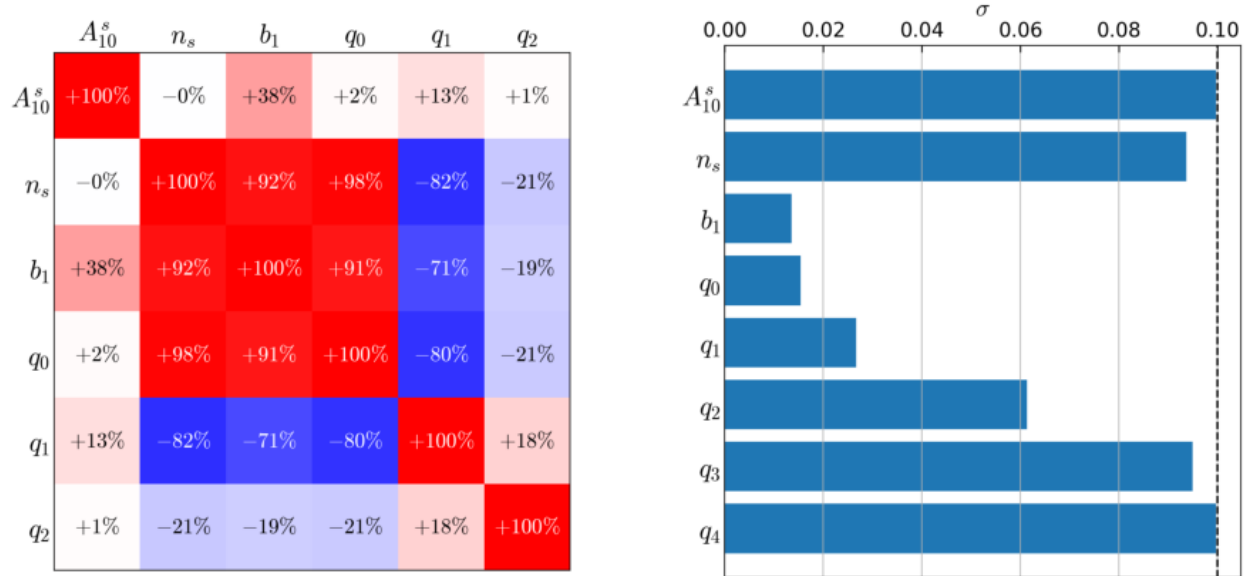
Authors: N. G. Karaçaylı, M. Ivanov, R. de Belsunce, C. Ravoux, J. M. Sexton, Z. Luki

Published: 2026-04-10T15:20:40+00:00

PDF: <https://arxiv.org/pdf/2604.09407v1>

Summary: The 1D flux power spectrum (P_{1D}) of the Ly α forest provides an exceptionally high-resolution probe of structure formation down to small scales ($k \approx 1 - 10 \ h^{-1} \text{Mpc}^{-1}$). These scales carry the imprints of massive neutrinos, warm dark matter, and the running of the primordial power spectrum spectral index. The effective field theory (EFT) is a promising perturbative approach to systematically and efficiently describe the Ly α forest, but it faces challenges in its application to P_{1D} , as many EFT parameters become degenerate when projected along the line of sight. In addition, this projection generates new stochastic terms from the integration over small-scale modes. In this work, we address these issues by compressing the EFT model space using the Fisher matrix formalism and linearizing the resulting compression directions, enabling analytic template marginalization and significantly reducing the computational cost of likelihood evaluation. We use hydrodynamical simulations to obtain a baseline estimate of EFT parameters, and use the DESI DR1 P_{1D} measurements to derive compression directions. We then marginalize over deviations from the baseline using these compression directions and forecast the constraining power of our formalism. We find that even in conservative scenarios where each data redshift bin requires its own set of EFT parameters, the cosmological constraints saturate with the linear bias, two leading-order 1D stochastic terms, and three principal combinations of the remaining EFT templates. In this case, our forecasted precision of the amplitude (Δ_p^2) and the logarithmic slope (n_p) of the linear matter power spectrum at the pivot scale ($k_p = 0.7 \text{ Mpc}^{-1}$) is 10% and 2.0%, respectively, which is similar to emulator-based analyses that include observational data systematics.

Selected figures



4. Constraining the Molecular Kennicutt-Schmidt Relation with Multi-Transition CO Observations of Nearby Galaxies

Focus: galaxies

Authors: Victoria G. G. Samboco, Ryan P. Keenan

Published: 2026-04-10T14:24:36+00:00

PDF: <https://arxiv.org/pdf/2604.09353v1>

Summary: The relationship between the star formation rate surface density and the molecular gas surface density in galaxies is key to understanding galaxy evolution. To investigate the molecular Kennicutt-Schmidt (K-S) relation and its dependence on gas density, we analyze a uniform sample of 36 nearby galaxies from the AMISS survey, focusing on the CO(1-0), CO(2-1), and CO(3-2) transitions, which trace progressively denser and warmer molecular gas. Using statistical methods that combine binning with Markov Chain Monte Carlo (MCMC) fitting, we derive the slope, scatter, and intercept of the SFR-CO relation for each transition. We find power-law slopes of 1.26, 1.14, and 1.07 for CO(1-0), CO(2-1), and CO(3-2), respectively, consistent with a trend toward increasingly linear star formation relations at higher-J transitions. This behavior supports the idea that denser gas is more directly linked to ongoing star formation and is consistent with previous findings of near-linear correlations between HCN or high-J CO luminosities and global SFR. The observed trend suggests an underlying relation between gas and SFR volume densities with a power-law index of ~ 1.5 , indicating enhanced star formation efficiency in denser environments. These findings underscore the critical role of dense gas in regulating star formation and highlight the importance of tracer selection and excitation conditions when interpreting the K-S relation across different environments.

5. A first [CII] view of high- z quiescent galaxies

Focus: galaxies

Authors: C. D'Eugenio, E. Daddi, R. Gobat, S. Jin, D. Liu, H. Sun, F. Gentile, F. Bruckmann, Z. Liu, I. Delvecchio, L. Vallini, B. Magnelli, A. Zanella

Published: 2026-04-10T14:19:03+00:00

PDF: <https://arxiv.org/pdf/2604.09347v1>

Summary: We present ALMA detections (or stringent upper limits) of the [CII] 158 μm emission line and underlying dust continuum from five massive quenched galaxies (QGs) at $2 < z < 4.7$. We find extreme variations in the molecular gas fractions ($f_g = M_{\text{mol}}/M_*$), spanning 0.1%-25%, if a standard $\alpha_{\text{[CII]}}$ applies. We attempt a first empirical calibration of $\alpha_{\text{[CII]}}$ with respect to dust continuum in a $z = 2$ lensed QG and with respect to CO(3-2) in a $z = 3.1$ QG, finding no evidence of strong deviations from the standard value. Dust continuum measurements, coupled with JWST/MIRI fluxes, suggest higher dust temperatures compared to expectations from $z < 2$ QGs, reaching $T_d \sim 40 - 50 \text{ K}$ in two galaxies. Coupled with remarkably high total infrared luminosities (LIR) not explained by observed JWST colors not by energy balance based on literature dust extinction measurements, and with [CII] deficits down to $[\text{CII}]/\text{LIR} \sim 2 \times 10^{-4}$ typical of (Ultra)Luminous Infrared Galaxies, our findings point to additional dust-heating mechanisms other than dust-absorbed stellar radiation. Surprisingly, JWST/NIRCam and ALMA imaging reveal widespread disturbed stellar morphologies and offsets/tails in dust and gas, indicative of ongoing interactions. While larger samples are needed to assess how common these features are in high- z QGs, these findings support a merger-driven origin for the phenomenology observed in these systems, with key similarities with respect to local post-starburst galaxies where low-velocity shocks and turbulence also inject energy into the residual ISM.

Selected figures

